

Foundation-Model Fusion in Lung Cancer Subtyping

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1 Introduction & Objectives

Accurate classification, subtyping, and grading of lung adenocarcinoma are essential for informing diagnosis and treatment decisions [1].

This project aims to predict the histologic growth pattern of lung adenocarcinoma from whole-slide images (WSIs) by integrating image embeddings generated by three pathology foundation models (FMs) with patient age and sex.

2 Data Preparation

The CLWD lung adenocarcinoma WSI and clinical dataset is publicly available in two formats [1]: a compressed JPG version [2] and a full-resolution SVS version [3]. The characteristics of the publicly available dataset are summarized in Table 1.

Table 1: Overview of the CLWD lung adenocarcinoma dataset.

Property	Value
Whole-slide images	408
Patients	210
Slides per patient	1–5 (median: 2)
Scan magnification	80×
Format	JPG and SVS
Disease	Lung adenocarcinoma
Label	Histologic growth pattern (7 classes)

2.1 Patient and Slide Selection

For this implementation, one slide per patient was selected using a deterministic rule, namely selecting the slide with the lowest `WSI_ID`. However, several exceptions required retaining multiple slides from the same patient.

- There are 209 distinct patient identifiers for 210 patients. Patient ID **8377886** occurs twice, with ages 68 and 69, corresponding to slides `WSI-35/36` and `WSI-103/104`, respectively. These records are treated as belonging to a single patient despite the samples being collected one year apart. One sample was selected from each age group (68 and 69 years).

Although these samples were collected one year apart, they may share patient-specific characteristics. This may introduce patient-level bias or allow the model to learn patient-specific patterns. Therefore, both samples were considered to originate from the same patient and were assigned to the same dataset split to prevent information leakage between the training, validation, and test sets.

- Five patients have conflicting histologic subtype labels across their slides: **8225322**, **8240634**, **8245975**, **8248415**, and **8248805**. Both slides were retained with their respective labels. However, all slides from an individual patient were treated as belonging to the same patient and were **not** distributed across different training, validation, or test splits.

2.2 Data Stratification and Cross-Validation

The dataset was divided into training (70%), validation (10%), and test (20%) sets using the 209 unique patient identifiers. Three folds were generated to perform cross-validation across three distinct patient-level data splits. The data distribution is shown in Table 2

Table 2: Patient-level stratification and WSI counts across the three folds. Total 215 WSIs from 209 Patients.

Split	Percentage	Patients	Fold 0 WSIs	Fold 1 WSIs	Fold 2 WSIs
Train	70%	146	149	149	151
Validation	10%	21	23	21	21
Test	20%	42	43	45	43

2.3 Downloading the WSI Data

Two publicly available versions of the dataset were considered: compressed JPG images and WSIs in SVS format. Reliable WSI processing requires metadata including objective magnification, microns per pixel (MPP), and image resolution at the highest-resolution level.

Although the JPG files are substantially smaller, they do not retain the required acquisition metadata. They also appear incomplete with respect to both the available metadata and the number of samples. Comparison between JPG files and their corresponding SVS files further suggests that the JPG images may not represent the full-resolution images acquired at the reported $80\times$ magnification.

This limitation prevents reliable determination and reproduction of a target magnification. For example, extracting images at $20\times$ from an $80\times$ scan requires four-fold downsampling. Without reliable metadata describing the original magnification, MPP, and image resolution, extraction at a specified target magnification cannot be consistently ensured.

For example, Table 3 compares WSI-1.svs with WSI-1.jpg.

Table 3: Comparison of available metadata for WSI-1.svs and WSI-1.jpg.

Metadata	WSI-1.svs	WSI-1.jpg
Vendor	Aperio	N/A
Highest resolution	185472×126336	64999×44275
Objective power (magnification)	$80\times$	N/A
MPP	$0.1038319 \mu\text{m}/\text{pixel}$	N/A
Level count	8	N/A

Because the highest-resolution levels of the SVS and JPG files do not correspond, it is not appropriate to assume that the JPG files represent full-resolution images scanned at $80\times$ magnification. Moreover, in the absence of acquisition metadata for the JPG images and corresponding SVS metadata, their objective power and physical resolution cannot be reliably determined. Therefore, SVS images were selected for this project because they provide the metadata required for informed, consistent, and reproducible image-processing and magnification-selection procedures.

3 Methodology

3.1 Step 1 — Feature extraction with three foundation models

3.1.1 Patch encoding

WSIs were processed using a sequential patch-based feature-extraction pipeline, as illustrated in Figure 1. Because WSIs are gigapixel-scale images, direct end-to-end processing is computationally impractical. Therefore, tissue-containing regions were first identified using Otsu-based segmentation at the thumbnail level, producing a tissue mask that excluded non-informative background regions.

Candidate patch locations were identified from the tissue mask at pyramid level 6 and mapped to level 0 coordinates using a scaling factor of 64. Image patches of size 224×224 pixels were subsequently

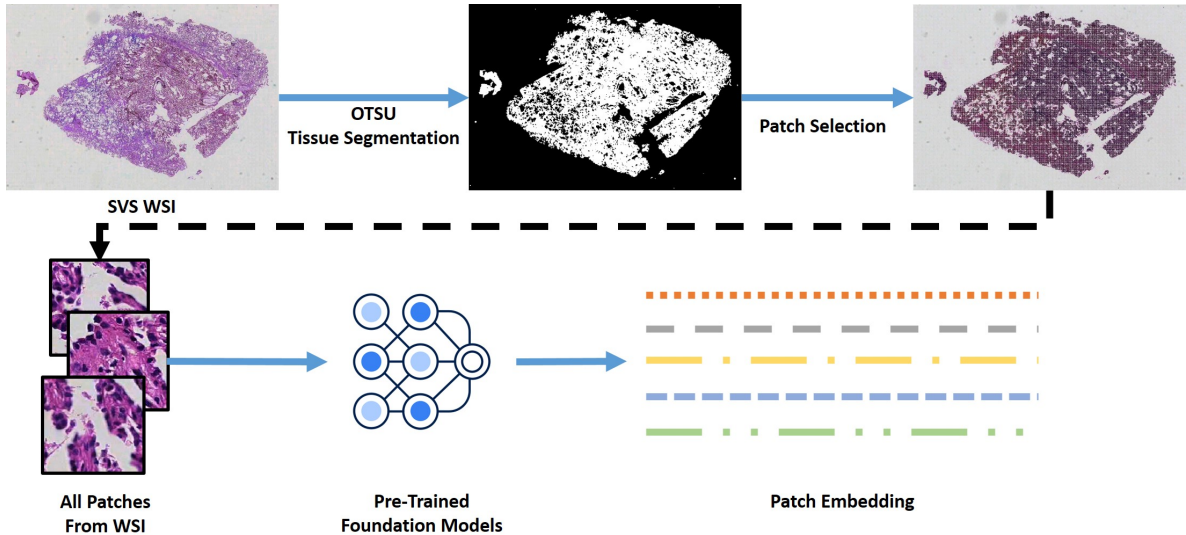


Figure 1: WSI processing and patch-embedding generation pipeline.

extracted from pyramid level 2, corresponding to $20\times$ magnification, using the `OpenSlide` library. A sliding-window stride of 224 pixels produced non-overlapping patches. Only patches with tissue coverage of at least 80% were retained, thereby reducing the inclusion of background and low-information regions.

The retained tissue patches were processed independently by three pretrained histopathology FMs: UNI2-h [4], Virchow2 [5], and H-OPTIMUS-1 [6]. These patch-level encoders generated feature embeddings that served as input representations for downstream slide-level aggregation and tumour subtype classification. The models were selected because of their large-scale pretraining on diverse WSI collections and their demonstrated performance across computational pathology tasks. Prism2 was excluded from patch-level feature extraction because it is designed as a slide-level encoder rather than a patch-level encoder. Table 4 summarizes the parameter count, embedding dimensionality, and average processing time per WSI for each model.

Table 4: Histopathology patch encoders used for feature extraction.

Model	Parameters (M)	Embedding dimension	Mean time/WSI (s)
UNI2-h	681	1536	722
Virchow2	632	1280	257
H-OPTIMUS-1	1100	1536	896

Each FM is used with the preprocessing recommended by its authors. UNI2-h and Virchow2 use channel-wise means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225); H-OPTIMUS-1 uses means (0.707223, 0.578729, 0.703617) and standard deviations (0.211883, 0.230117, 0.177517).

3.1.2 Slide encoding and classification

After patch-level feature extraction, embeddings from each WSI are aggregated into a slide-level representation for classification into one of seven histopathological tumour classes. An attention-based multiple instance learning (ABMIL) [7] framework treats each WSI as a bag of patch-level instances, with the slide-level tumour subtype as the bag-level label. The attention mechanism learns the relative importance of patches and emphasizes diagnostically informative regions in the aggregated slide representation.

ABMIL was selected instead of conventional mean pooling or a pretrained slide-level encoder for several reasons. Mean pooling is computationally efficient but gives every patch equal weight. When tumour tissue occupies only a small portion of a WSI, patches from non-tumour regions, such as

connective or adjacent normal tissue, may dominate the pooled representation and dilute tumour-associated features. Attention-based aggregation instead learns instance-specific weights that can prioritize patches informative for slide-level classification.

Pretrained slide-level encoders present another consideration. Prism2 was fine-tuned with Virchow2 patch embeddings and is designed for that feature space. Applying it directly to embeddings from UNI2-h or H-OPTIMUS-1 could introduce a feature-space mismatch: Virchow2 yields 1280-dimensional patch embeddings, whereas UNI2-h and H-OPTIMUS-1 each yield 1536-dimensional embeddings. Prism2 would therefore not provide a consistent, model-independent aggregation strategy across all three models.

ABMIL can be trained separately on each FM’s patch embeddings while retaining the same aggregation architecture. Its projection layer maps the resulting slide-level representations to a common 1024-dimensional space. This both permits attention-weighted aggregation of variable numbers of patches and facilitates subsequent fusion of slide representations from UNI2-h, Virchow2, and H-OPTIMUS-1 without an additional dimensionality-alignment step. Figure 2 shows the ABMIL slide-aggregation framework.

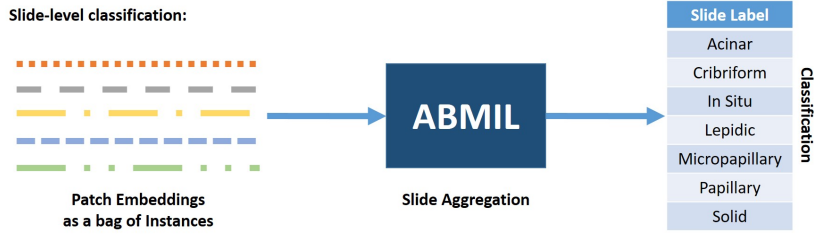


Figure 2: Attention-based multiple instance learning for slide aggregation.

3.2 Clinical metadata encoding and classification

Patient age and sex are incorporated as clinical features for predicting the seven histopathological tumour subtypes. The clinical-feature classifier is trained in two stages. First, an MLP receives age and sex as inputs and is trained to reconstruct the corresponding clinical features, with the aim of learning a general representation of the metadata. Second, the representation-learning layers are frozen, and a classification head is added to predict the tumour subtype from the learned representation.

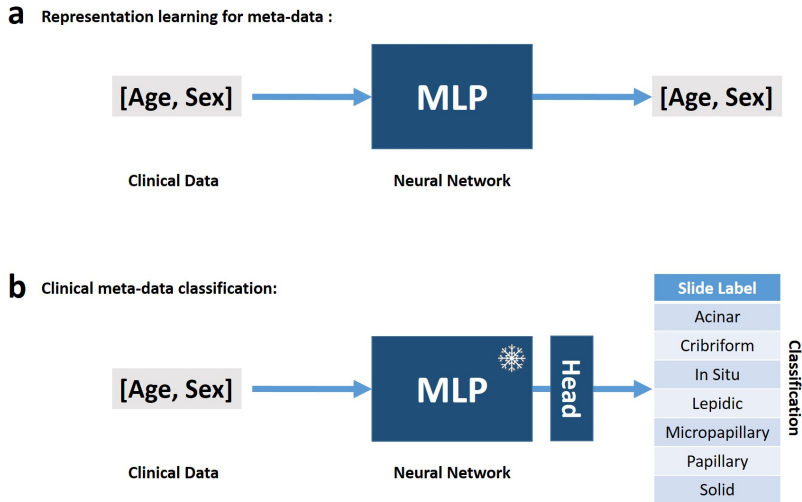


Figure 3: Two-stage clinical metadata encoding and tumour-subtype classification.

Sex is binary encoded (female: 0; male: 1). Age is min-max normalized using the minimum and maximum ages in the dataset, 24 and 80 years, respectively, mapping values to the interval $[0, 1]$. This

places age on a comparable numerical scale for optimization. The MLP takes a two-dimensional vector containing normalized age and encoded sex and ultimately produces a probability distribution over seven tumour subtypes. Figure 3 illustrates the two-stage classifier.

3.3 MIL and clinical metadata fusion (manual)

To improve slide-level classification, the best-performing MIL model identified during slide-level (AB-MIL) evaluation is combined with patient age and sex. The selected MIL aggregator is frozen, and its best-performing checkpoint generates a 1024-dimensional embedding for each slide.

In parallel, normalized age and binary-encoded sex are supplied to the frozen, previously trained MLP-based clinical encoder (shown in Fig 3 a), producing a 384-dimensional representation for each patient.

The slide and clinical representations are concatenated into a joint multimodal feature vector and passed to a trainable MLP fusion and classification head. The network learns interactions between histomorphological and clinical representations and predicts one of seven tumour subtypes. The architecture comprises a frozen MIL-based slide encoder, a frozen MLP-based clinical encoder, and a trainable MLP fusion head, as illustrated in Figure 4.

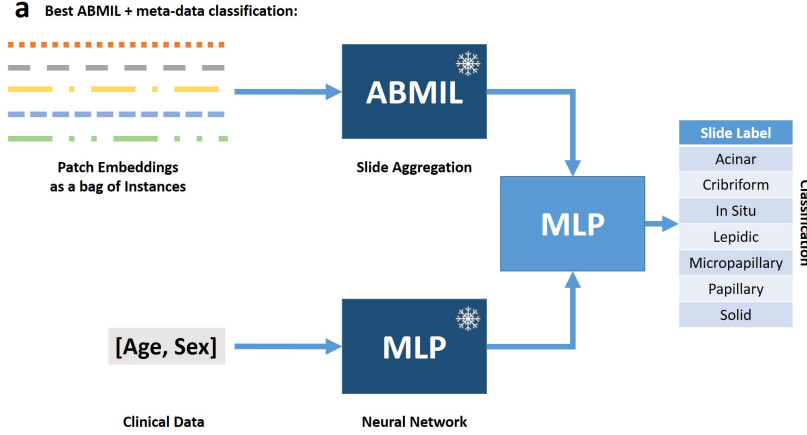


Figure 4: Fusion of slide-level and clinical metadata representations for tumour-subtype classification.

3.4 Task 2 — Agent-Guided Foundation-Model and clinical metadata Fusion

3.4.1 Objective and input representations

An automated fusion agent searches for an effective combination of three pathology FM slide embeddings and patient age and sex to predict one of seven lung adenocarcinoma histologic growth-pattern classes. Slide-level representations are generated from UNI2-h, Virchow2, and H-OPTIMUS-1 patch embeddings using trained ABMIL aggregators and cached before the fusion search. Caching avoids repeated FM inference and MIL aggregation during candidate evaluation. Age is normalized, and sex is binary encoded; a separately learned clinical encoder can provide a metadata representation.

For fold f , the resulting dataset contains N_f WSI samples and the following arrays:

$$\begin{aligned} \mathbf{X}_{\text{UNI2}} &\in \mathbb{R}^{N_f \times 1024}, & \mathbf{X}_{\text{Virchow2}} &\in \mathbb{R}^{N_f \times 1024}, \\ \mathbf{X}_{\text{H-OPTIMUS-1}} &\in \mathbb{R}^{N_f \times 1024}, & \mathbf{X}_{\text{clinical}} &\in \mathbb{R}^{N_f \times 384}, \\ \mathbf{y} &\in \{1, \dots, 7\}^{N_f}, & \mathbf{wsi_id} &\in \mathcal{I}^{N_f}. \end{aligned}$$

Here N_f is the number of WSI samples in the fold, $\mathbf{y}$ contains subtype labels, and $\mathbf{wsi_id}$ identifies the slides. Multiple slides belonging to the same patient remain in the same data partition.

3.4.2 Fusion search space

The agent considers three model families:

1. **Early concatenation:** Optionally L2-normalize each pathology stream and project it into a smaller learned space; transform the clinical inputs into a learned representation; concatenate the representations; and train a fusion MLP followed by a seven-class classifier. Normalization addresses possible scale differences between FM features, while projection can reduce parameter count in the small cohort.
2. **Late fusion:** Train stream-specific classifiers and combine their outputs by mean probability, mean logit, validation-weighted voting, or stacking. Stacking uses out-of-fold base-model predictions for the training patients to avoid fitting its regularized multinomial logistic-regression meta-classifier on in-sample predictions.
3. **Gated fusion:** Project all four streams into a shared latent space. A gate network produces four patient-specific softmax weights; their weighted sum is passed through a fusion MLP and a seven-class classifier.

3.4.3 Sequential search and model selection

The search begins with coverage trials spanning early concatenation without normalization, with normalization, and with normalization plus projection; the four late-fusion variants; and gated fusion. Once all families have been assessed, the agent proposes targeted trials based on the leading validation result: projection and dropout variants for early fusion, stacking or gating for late fusion, or reduced shared dimension, stronger dropout, and another seed for gated fusion. Each proposal includes a textual rationale and is recorded in a JSON file decision log.

For neural candidates, model weights are restored from the epoch with the highest validation balanced accuracy. Candidate predictions are converted to class assignments by $\hat{y}_i = \arg \max_c p_{ic}$, and trial selection primarily uses validation balanced accuracy. Each checkpointable neural trial stores its strategy, full configuration, input dimensions, class count, validation score, state dictionary, and proposal metadata. If a neural trial improves the best validation score, its checkpoint is updated with the best model configuration.

The search uses a fixed trial budget, requires coverage before adaptation, applies a minimum material-improvement threshold and patience-based stopping, fixes random seeds, and does not use test data for trial selection. The selected architecture is reconstructed from the saved strategy and configuration for final test inference using the test set.

Table 5: Reported training settings. AdamW was used in all four settings; class-weighted cross-entropy weights were derived from the training partition.

Setting	ABMIL	Metadata only	Best ABMIL + metadata	Fusion agent
Learning rate	10^{-4}	10^{-3}	10^{-4}	10^{-4}
Weight decay	10^{-4}	0	10^{-4}	10^{-4}
LR decay	$\gamma = 0.1$ every 3 iterations	$\gamma = 0.1$ every 10 iterations	$\gamma = 0.1$ every 3 iterations	$\gamma = 0.1$ every 10 iterations
Minimum LR	10^{-7}	10^{-7}	10^{-7}	10^{-7}
Epochs	10	200	100	100
Batch size	1	1	1	16
Early-stop patience	Not specified	50 epochs	50 epochs	50 epochs
Loss	Weighted cross-entropy	MSE for representation learning; weighted cross-entropy for classification	Weighted cross-entropy	Weighted cross-entropy
Checkpoint selection	Validation loss or balanced accuracy, as configured	Validation balanced accuracy	Validation balanced accuracy	Validation balanced accuracy

4 Results

All experiments were performed on the Windows platform using Python 3.11, PyTorch 2.9, and CUDA 12.8. The system was equipped with an Intel Core i9-12900K processor 64 GB of RAM, and NVIDIA RTX 3090 GPU with 24 GB of VRAM.

4.1 Experimental configuration

Patch embedding generation used the preprocessing and inference settings recommended for each FM. For each fold, training samples optimized model weights, validation samples selected checkpoints and fusion configurations, and held-out test samples were reserved for final evaluation. The same ABMIL configuration was applied to each FM so that slide-level differences could be compared under a consistent aggregation procedure. Table 5 summarizes the reported settings.

4.2 Three-fold test performance

Mean performance across three held-out test folds was summarized. Table 6 reports the provided percentages as mean $\pm$ standard deviation and 95% confidence intervals (CIs).

Table 6: Seven-class subtype-classification performance across three test folds (percent). Values are mean $\pm$ standard deviation [reported 95% CI].

Configuration	Macro AUROC (%)	Balanced accuracy (%)
Metadata only (age + sex)	61.08 ± 3.04 [53.50, 68.65]	23.95 ± 2.10 [18.73, 29.17]
UNI2-h + ABMIL	86.75 ± 3.34 [78.45, 95.06]	51.82 ± 2.65 [45.22, 58.42]
Virchow2 + ABMIL	85.32 ± 3.78 [75.92, 94.71]	52.86 ± 8.19 [32.51, 73.21]
H-OPTIMUS-1 + ABMIL	88.24 ± 2.19 [82.79, 93.69]	61.81 ± 4.91 [49.60, 74.02]
H-OPTIMUS-1 + ABMIL + metadata	88.06 ± 3.45 [79.48, 96.64]	62.13 ± 1.90 [57.39, 66.86]
Agent-selected fusion + metadata	88.30 ± 3.17 [80.42, 96.18]	63.43 ± 4.81 [51.46, 75.39]

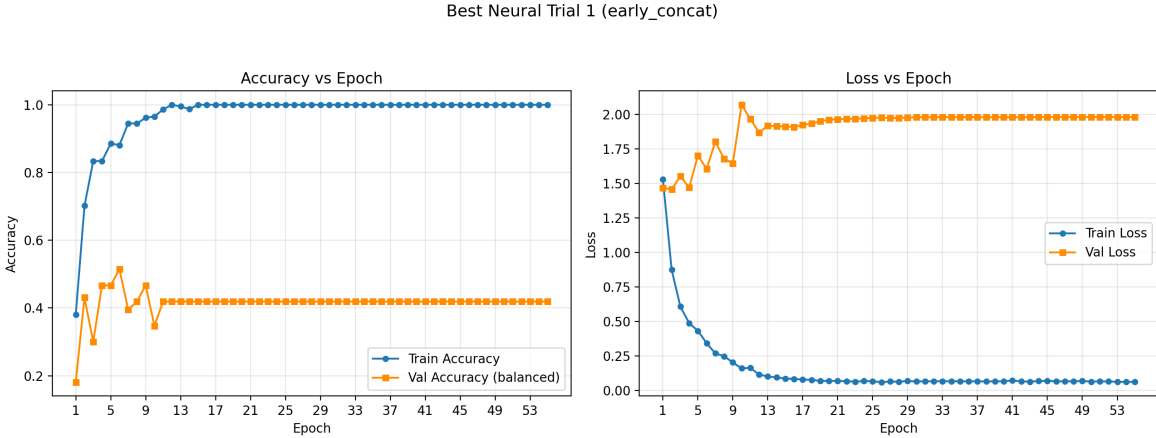


Figure 5: Training and validation curves for the agent-selected fusion model on the Fold 2 training and validation sets.

Detailed per-fold results, including confusion matrices, class-wise receiver operating characteristic (ROC) curves, classification reports, sensitivity, and specificity, are provided in the [results directory](#). Checkpoints and training curves for the non-agentic models are available in the [checkpoints directory](#). For non-agentic models, H-OPTIMUS-1 (ABMIL) Fold 2 shows the best performance (balanced accuracy of 66.77% and macro AUROC of 91.50%). Among the agentic-fusion models, Fold 2 achieved its highest reported test result with a balanced accuracy of 68.70% and macro AUROC of 92.50%. Agent fusion selected strategy was early concatenation without L2 normalization or feature projection, with hidden dimension 384, dropout 0.3, learning rate 10^{-4} , weight decay 10^{-4} , 100 epochs, batch size 16,

and seed 42. Figure 5 illustrates the training and validation loss and balanced accuracy across epochs. Although a substantial divergence is observed between the training and validation curves, model selection was based exclusively on the epoch achieving the highest validation balanced accuracy. This checkpointing strategy reduces the likelihood of selecting a model that has overfit the training data. Figures 6 and 7 display the supplied Fold 2 evaluation graphics.

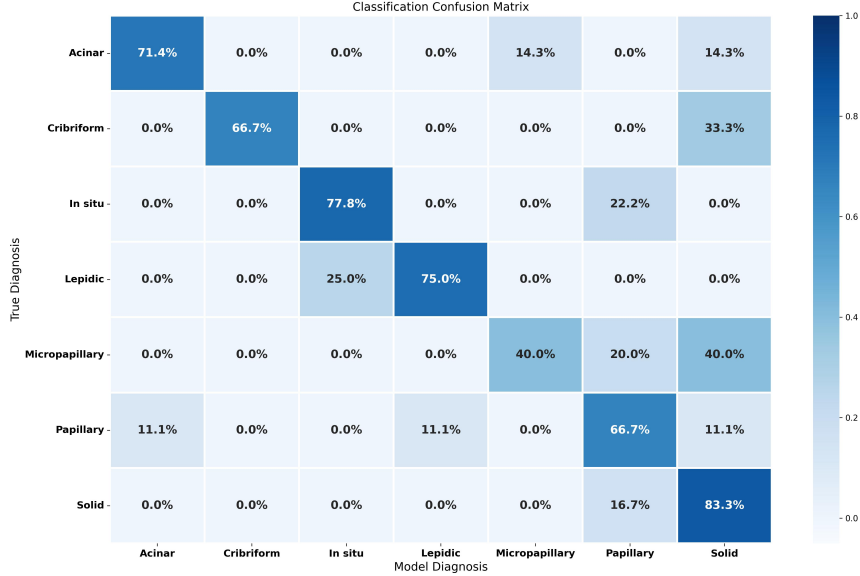


Figure 6: Confusion matrix for the agent-selected fusion model on the Fold 2 test partition.

5 Discussion

Classifying seven histologic growth patterns from WSIs is complicated by morphologic heterogeneity, overlapping features between subtypes, and large non-tumour regions. The metadata-only classifier achieved 23.95% mean balanced accuracy and 61.08% mean macro AUROC, indicating limited discrimination from age and sex alone. In contrast, all three FM-ABMIL pipelines performed better; H-OPTIMUS-1 + ABMIL was the strongest single-FM configuration, achieving 61.81% balanced accuracy and 88.24% macro AUROC.

Adding age and sex to the H-OPTIMUS-1 representation produced a mean balanced accuracy of 62.13% and macro AUROC of 88.06%. The corresponding fold-wise balanced accuracies were 60.20%, 62.18%, and 64.01%, compared with 61.73%, 56.94%, and 66.77% for H-OPTIMUS-1 + ABMIL alone. Thus, manual metadata fusion improved Fold 1 but not Folds 0 or 2, and its mean AUROC was similar to the slide-only baseline.

The agent-selected fusion approach yielded fold-wise balanced accuracies of 62.33%, 59.26%, and 68.70%, with a mean of 63.43% and macro AUROC of 88.30%. Relative to manual H-OPTIMUS-1 + metadata fusion, it improved balanced accuracy in Folds 0 and 2 but decreased it in Fold 1. Unlike manual fusion, which uses the strongest single MIL stream, the agent searches combinations of all three FM-derived slide embeddings and metadata. Different strategies were selected across folds, suggesting sensitivity to data partitioning. The small mean gain over the strongest single-FM baseline and variability across folds do not establish a consistent benefit from greater fusion complexity; computational and inference costs should also be considered.

6 Failure Analysis

The metadata-only model showed weak apparent discrimination for Acinar, Cribriform, and Lepidic classes, with more pronounced prediction patterns for In Situ and Solid. These are descriptive observations from the reported confusion matrices rather than evidence of causal demographic associations.

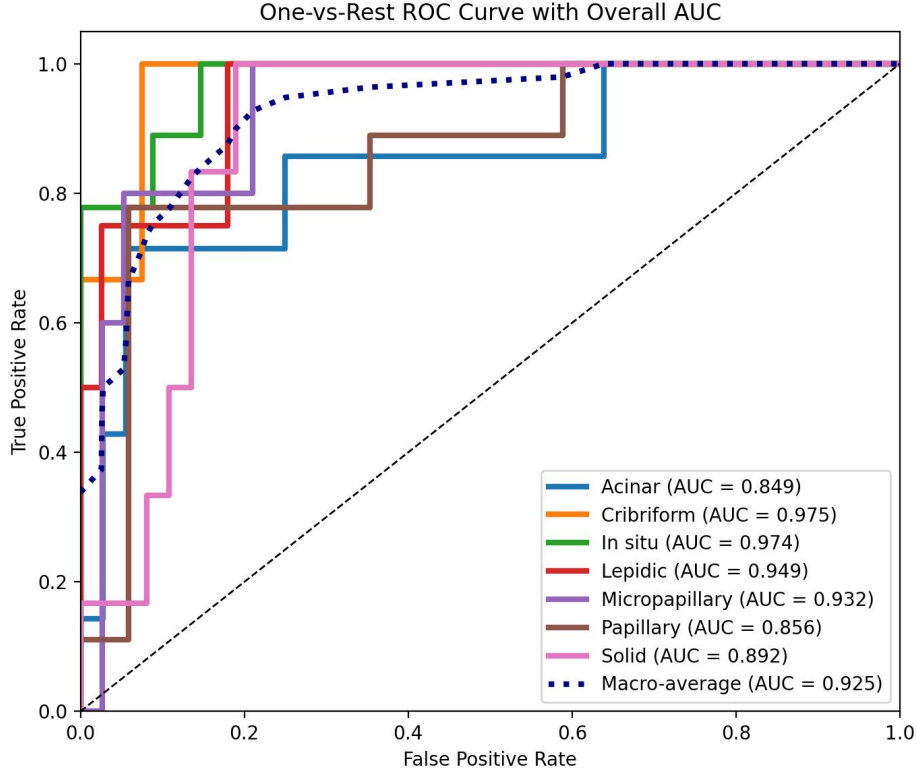


Figure 7: Class-wise ROC curves for the agent-selected fusion model on the Fold 2 test partition.

With H-OPTIMUS-1 + ABMIL, errors varied by fold. In Fold 0, Acinar was often predicted as Papillary and Micropapillary as Cribriform. In Fold 1, Acinar was confused with Papillary or Lepidic, and Micropapillary with Cribriform. In Fold 2, Papillary was confused with Micropapillary, while Acinar was variously confused with Solid, Papillary, and Cribriform. These recurrent errors are consistent with overlapping learned subtype representations, although confusion matrices alone cannot establish the underlying morphologic cause.

Adding metadata changed rather than uniformly reduced these errors. In Fold 0, Acinar was confused with Lepidic and Cribriform with Solid and other classes. In Fold 1, Cribriform was often predicted as In Situ or Solid, Micropapillary as Cribriform, and Papillary as Acinar, Micropapillary, or Lepidic. In Fold 2, Acinar and Solid were confused with Cribriform and Micropapillary, and Lepidic with In Situ. The impact of metadata therefore appears partition- and subtype-dependent.

7 Future Work

With approximately tenfold greater computational resources, future experiments could adapt self-supervised FMs to target-domain lung adenocarcinoma WSIs rather than relying solely on frozen pretrained embeddings. Additional clinical and pathological variables, if available, could be evaluated for incremental value beyond age and sex. Multi-scale modeling could combine fine cellular and nuclear detail at higher magnification (for example, 40 $\times$) with the larger architectural context needed to identify growth patterns. Candidate methods include multi-scale patch extraction, hierarchical aggregation, and joint high-resolution/local and low-resolution/contextual representations. These extensions should be assessed against single-modality baselines, taking account of validation stability, test performance, computational expense, and inference complexity.

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